BI2025 Experiment Report - Group 84

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Abstract

This report documents the machine learning experiment for Group 84, following the CRISP-DM process model. The primary business objective was to develop a supportive tool for salary adaptation, specifically to assist in identifying existing employees who may be over- or under-compensated, as well as to benchmark appropriate salaries for new hires. Leveraging data from the Stack Overflow 2025 Annual Developer Survey, we formulated a linear regression task to predict developer salaries. While modern salary prediction often utilizes complex non-linear algorithms, we deliberately selected a Ridge Regression model to prioritize interpretability. Although the inherent variance in self-reported compensation data prevented reaching the ambitious initial target of $R^2 > 0.80$, our model achieved an R^2 of 0.57. This result is comparable to current state-of-the-art benchmarks established on similar datasets (e.g., 2024), validating the linear approach as a robust, transparent baseline for decision support.

CCS Concepts

• **Computing methodologies → Machine learning**.

Keywords

CRISP-DM, Provenance, Knowledge Graph, Machine Learning, Linear Regression, Salary Prediction, Stack Overflow Developer Survey 2025,

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1 Business Understanding

The Business Understanding phase serves as the foundational step of the CRISP-DM methodology. In this chapter, we outline the project's scope by defining the specific business problem, the available data, and the criteria for success as well as potential AI risk aspects.

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1.1 Data Source and Scenario

We founded the company "Fritz & Maier AG" after our successfully completed Bachelors. It is a software company that has scaled rapidly to hundreds of employees located all over the world. Due to this rapid expansion, the current compensation structure is ad-hoc, resulting in internal pay inequities and potential misalignment with the current market. As the company matures and grows even further, HR requires a data-driven approach to standardize these salaries. The goal is to use the Stack Overflow data to analyze the relationship between skills, experience, and education against compensation to benchmark current employee salaries and make adaptations for the following business year. To achieve this, the analysis utilizes the Stack Overflow Annual Developer Survey 2025. This dataset contains comprehensive information regarding the developers background.

1.2 Business Objectives

1.2.1 Objective 1. The primary objective is to establish a data-driven salary benchmarking model to find a fair market value for software engineering roles. This model must calculate compensation based on objective factors such as professional experience, geographical location, and specific technology stacks, rather than subjective negotiation.

1.2.2 Objective 2. Our second objective focuses on internal equity. We aim to identify current employees who may be significantly under- or overpaid compared to the market. This identification allows HR to proactively address retention risks or budget inefficiencies.

1.2.3 Objective 3. The third objective is to standardize salary offers for new hires. By having a reliable model, we aim to reduce the time spent on negotiations and create a more transparent hiring process that instills confidence in candidates.

1.2.4 Objective 4. The fourth objective is budget optimization. We aim to ensure the company's financial resources are utilized efficiently by avoiding arbitrary overpayment while maintaining a competitive stance in the talent market.

1.3 Business Success Criteria

1.3.1 Criterion 1. The first criterion addresses the adoption of the model. The project will be considered successful if the HR department adopts the model to review at least 80% of current employee contracts.

1.3.2 Criterion 2. The second criterion focuses on retention. We aim to see a measurable reduction in staff turnover that is specifically attributed to compensation issues.

1.3.3 Criterion 3. The third criterion regards recruitment efficiency. Success is defined as an increase in the acceptance rate of initial job offers and a reduction in the time-to-hire, proving that our data-backed offers are competitive.

1.3.4 Criterion 4. The fourth criterion is financial accuracy. The project is successful if we achieve a total salary budget that aligns within +/- 10% of the predicted market benchmark, ensuring the company is neither overspending nor significantly underpaying.

1.4 Data mining goals

1.4.1 Goal 1. The general overall goal of our data mining activities is to build a regression model which helps HR determine appropriate salaries for new employees. And in order to do that we take the dataset provided by the Developer survey, prepare the data accordingly and build the model out of it.

1.4.2 Goal 2. Our second data mining goal is to identify employees that are significantly over- / underpaid in comparison to the market. Here we want to compare the predicted salary with the actual salary and afterwards be able to flatten out deviations.

1.4.3 Goal 3. Our third goal is to identify which skills or attributes / features influence the salary prediction of a developer the most. For this we want to look at the features separately and then identify the most driving ones.

1.4.4 Goal 4. The fourth and last here explicitly listed data mining goal is to analyze the relationship between salary deviations and indicators of job satisfaction, especially regarding underpaid developers, in order to identify the risk of an employee leaving the company early .

1.5 Data mining success criteria

1.5.1 Criterion 1. The first and most important data mining success criterion is to achieve a high accuracy regarding the salary predictions. Therefore we want to reach an R^2 of 80% minimum.

1.5.2 Criterion 2. Our second goal is to have a robust model which on average minimizes the predictions with large absolute errors. We consider the salary of an employee to be a large error, if he has at least +/- 3000 euros salary deviation of the initial model prediction. In this regard we use the MAE in order to measure this performance overall of our model.

1.5.3 Criterion 3. Our third goal is to have a high data quality for our model, which means, that after all data cleansing we have less than 10% of missing data across all features.

1.5.4 Criterion 4. Our fourth goal is to make the model readable for the HR department. As this is a subjective criterion it has to be checked whether the employees of the HR department are able to properly read the model and take conclusions out of it. In order to guarantee that, the creators of the model validate the usage of the HR department.

1.6 AI Risk Aspect

1.6.1 Fairness & Ethics. A significant advantage of the chosen dataset is the absence of gender differentiation, which inherently lowers the risk of the model developing a gender-based bias. To further ensure reliability, we aim to mitigate the risks associated with noisy or missing data as much as possible during the preparation phase. However, we acknowledge that other potential biases remain, the most prominent being the country of employment, which could significantly influence the model's predictions and will require careful handling.

1.6.2 Data Quality. Regarding data quality, it is important to note that while the Stack Overflow survey covers a broad range of developers, it does not fully represent the global developer population or universal salary distributions. Additionally, since Fritz & Maier AG currently utilizes a unique internal methodology for setting salaries, there is a risk that the trained model may strongly misalign with current internal standards, potentially limiting its immediate applicability without calibration.

1.6.3 Documentation & Usage. In the sensitive domain of salary prediction, maintaining high standards of governance and documentation is critical. This is necessary not only to ensure compliance with regulations such as the EU AI-Act but also to inform the HR department of potential model limitations. Consequently, it is imperative that HR remains aware of the possibility of biased predictions and closely monitors or reviews every salary recommendation generated by the system.

2 Data Understanding

As one of the most comprehensive snapshots, the **Stack Overflow 2025 Annual Developer Survey** dataset provides a diverse range of features including geographic location, developer experience, technology stack, and employment details.

2.1 Attribute Types

Representing the survey's fifteenth year, it aggregates over 49,000 responses from 177 countries. The data includes responses to 62 questions regarding 314 different technologies, providing specific insights into the adoption of AI agent tools, Large Language Models (LLMs), and community platforms [13]. Complementing the primary response data, the dataset provides a metadata schema file. This file acts as a data dictionary, defining the question text and expected response format for variables in the main dataset. The primary data file contains 172 columns and 49,191 rows. The schema file consists of 6 columns and 139 rows, categorizing the survey questions into four distinct types, shown in Figure 1:

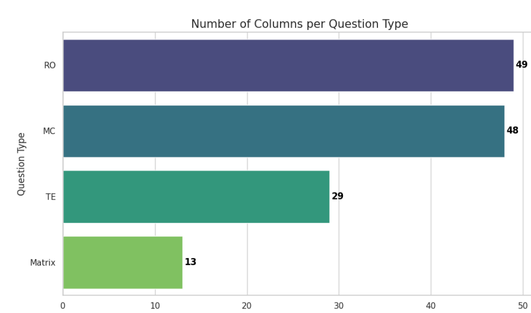

Figure 1: Listing of different question types with corresponding amount

As shown in Figure 1, the dataset comprises 48 Multiple Choice (MC), 49 Rank Options (RO), 29 Text Entry (TE), and 13 Matrix columns. Ideally, each row in the schema would correspond to a column in the response data, providing context for the recorded values. However, as indicated by the count difference - 139 schema definitions versus the 172 data columns - the schema does not cover every variable present in the dataset. This discrepancy, including missing definitions and column name mismatches, is addressed in detail in the next Section 2.1.1.

2.1.1 Schema Consistency and Column Mismatch. To validate the structural integrity of the dataset, we performed a programmatic comparison between the column names in the primary dataset and the `qname` identifiers in the schema. This was executed using set operations to calculate the intersection and relative complements of the variable lists. The automated consistency check yielded **126 Common Columns** that matched perfectly, **46 Data-Only Columns** that existed in the dataset but were missing from the schema, and **13 Schema-Only Columns** that were defined in the schema but absent from the dataset. Manual analysis was required to reconcile these differences. The investigation revealed that the mismatch is primarily due to the "flattening" of complex question types:

(1) **Matrix Question Expansion:** The 13 columns found only in the schema (e.g., `Language`, `Database`, `AIAgentImpact`) correspond to "Matrix" style questions. In the dataset, these are expanded into multiple columns representing specific aspects or Likert scale responses (e.g., `LanguageHaveWorkedWith`, `LanguageWantToWorkWith`, `AIAgentImpactStrongly agree`).
(2) **Derived Features:** Several columns found only in the data are calculated or system-generated fields rather than survey questions. Key examples include `ResponseId` (primary key) and `ConvertedCompYearly` (the target variable for salary prediction).

Due to this structural divergence, manual mapping confirmed that the missing columns were contextual expansions using a suffix-based convention: **-HaveWorkedWith** captures current usage, **-WantToWorkWith** denotes future interest, and **-Admired/Desired** reflects sentiment towards specific technologies. Consequently, for the modeling phase, features such as `LanguageHaveWorkedWith` were treated as the primary representation of the `Language` variable defined in the schema.

2.2 Statistical Properties and Feature Scoping

Given the dataset's high dimensionality (172 columns), we restricted the analysis to a subset of candidate features selected via domain reasoning, prioritizing variables with a logical theoretical link to compensation (e.g., *WorkExp*, *Country*). While we acknowledge that this heuristic approach introduces potential selection bias, it prioritizes interpretability and computational efficiency over exhaustive search. A specific challenge arose from "Matrix" style questions (e.g., `TechEndorse`, `JobSatPoints`), which expand into dozens of sparse binary features. To optimize dimensionality without discarding predictive signals, we applied a statistical filtering mechanism: for each high-cardinality ranking group, we calculated the Pearson correlation coefficient between every option and the target variable (`ConvertedCompYearly`), eventually retaining only the top three features with the highest absolute correlation per group.

2.2.1 Results. This process reduced the ranking feature count from 45 columns down to 9. Notably, the correlations were generally weak (absolute values < 0.05), suggesting that individual technology opinions are minor factors in salary determination. All other ranking columns were dropped from the analysis. This combined approach reduced the initial 172 columns to a final subset of **31 high-impact features** for the modeling phase, meaning a total of 141 columns were excluded from the analysis.

2.2.2 Feature Statistics and Distribution Analysis for selected Features. We analyzed the descriptive statistics for the 31 retained features to understand the central tendencies and variability of the dataset.

2.2.3 Demographic and Professional Profile. The "typical" respondent in this dataset fits a specific profile characterized by high professional engagement and a distinct demographic skew. The majority of the dataset (76%) consists of individuals identifying as professional developers who are currently employed. In terms of background, the modal age bracket is 25-34 years, with a Bachelor's degree being the most common educational attainment. Experience levels are significant, with the average respondent possessing approximately 16.5 years of total coding experience and 13.4 years of professional work experience. Both experience distributions are right-skewed (skew ≈ 1.2–1.5), indicating a long tail of highly experienced veterans. Geographically and professionally, the United States is the most frequently represented country, and "Developer, full-stack" appears as the most common job role.

2.2.4 Target Variable Analysis: Salary. The target variable, `ConvertedCompYearly`, exhibits extreme distributional characteristics that will pose challenges for linear modeling. The mean salary is $101,761, which is significantly higher than the mode of $69,609, indicating a strong right skew. This non-normality is numerically confirmed by a massive skewness of **75.5** and a kurtosis of **7059**. Furthermore, the maximum recorded salary reaches $50,000,000, a value likely attributable to data entry errors or currency conversion anomalies. These metrics implicate an extreme non-normality that necessitates robust outlier handling and likely a logarithmic transformation in the preparation phase. A more detailed outlier analysis is provided in Section 2.5.

Correlation Analysis

To rigorously assess the linear relationships between the selected numerical features and the target variable, we computed a Pearson correlation matrix. The results were visualized in a lower-triangle heatmap (see Figure 2), where the color intensity represents the strength of the relationship - red indicating positive correlation and blue indicating negative correlation.

2.2.5 Methodology. The analysis employed a programmatic "Smart Drop" logic, systematically flagging features with **Weak Signals** ($|r| < 0.05$) for low predictive value and resolving **Multicollinearity** ($|r| > 0.60$) by retaining only the variable in correlated pairs that exhibited the stronger relationship to the target.

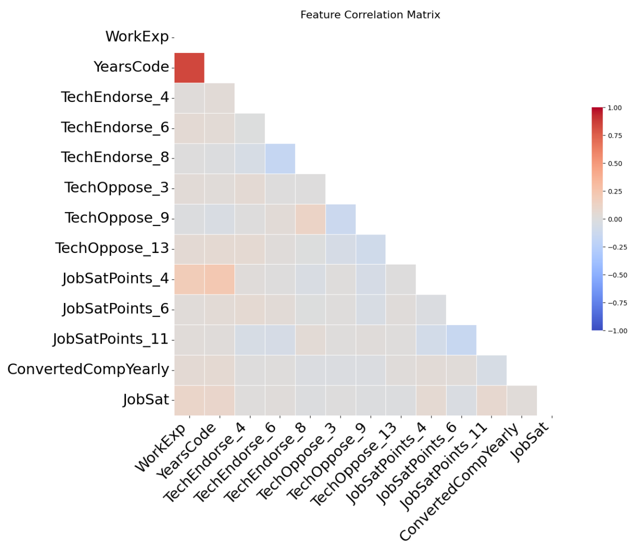

Figure 2: Pearson Correlation Matrix of selected numerical features. The deep red block highlights the strong multicollinearity between experience-related variables, while the faint bottom row indicates weak linear correlations with the target variable.

2.2.6 Visual Analysis and Results. The heatmap reveals a predominantly neutral field, indicating that most independent variables have low linear correlation with each other. This independence is generally favorable for regression models. However, two specific areas warrant closer inspection:

1. Feature-to-Feature Correlations (Multicollinearity). The most prominent visual feature is the deep red block in the top-left corner, representing the relationship between `YearsCode` and `WorkExp`. As expected, these variables are strongly positively correlated, exceeding the 0.60 collinearity threshold. There is an excpetion to the rule, as although our algorithm suggested removing one of these variables to reduce redundancy, empirical testing during the modeling phase revealed that retaining both yielded superior performance. This suggests that while they overlap, they capture distinct nuances. `YearsCode` accounts for total coding history (including education and hobbies), while `WorkExp` strictly measures professional tenure. Therefore, we overrode the automated suggestion and retained both features.

2. Correlations with the Target (ConvertedCompYearly). The row corresponding to the target variable (`ConvertedCompYearly`) is notably faint across the board, with most cells showing near-zero color intensity. This lack of strong linear correlation highlights a critical data characteristic: the relationship between developer attributes and salary is likely non-linear, or heavily obscured by the extreme outliers identified in the previous statistics section (e.g., salaries of $50M). This visual confirmation supports the necessity for the non-linear transformations (such as log-scaling) applied in the subsequent Data Preparation phase.

2.3 Data Quality Assessment

2.3.1 Missing Values Analysis. The dataset exhibits significant sparsity across several feature categories. Figure 3 illustrates the frequency of missing entries per column.

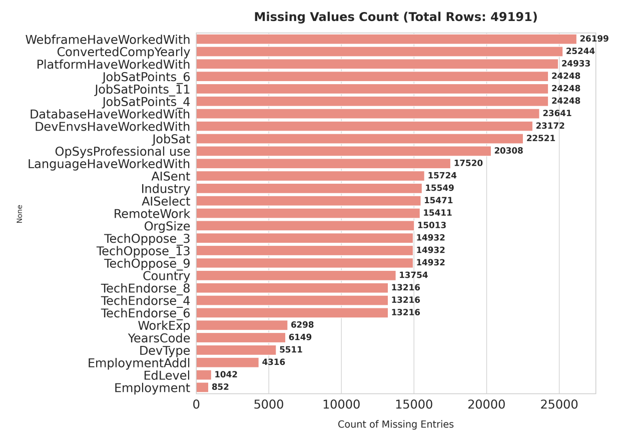

Figure 3: Count of missing entries per feature (Total Rows: 49,191). The target variable and detailed tech stack questions show high missingness.

After a detailed analysis of Figure 3, several critical quality issues were identified. The most severe is the absence of salary data in the target variable (`ConvertedCompYearly`), affecting 25,244 records (approximately 51% of the dataset); since supervised learning requires labeled data, these rows must be dropped during the preparation phase. Conversely, high missingness in tech stack and ranking columns, such as `WebframeHaveWorkedWith` (26,199 missing) and `JobSatPoints` (24,248 missing), likely indicates valid "Not Applicable" responses or survey fatigue rather than random error, necessitating imputation rather than deletion. Finally, core demographic fields like `EdLevel` and `Employment` exhibit very low missingness ($< 2\%$), providing a stable foundation for the analysis.

2.3.2 Plausibility. Beyond missing data, we validated the logical consistency of key numerical variables against defined logic thresholds to identify anomalies. Regarding the target variable, our analysis reveals extreme salary outliers that would severely distort a linear model such as Ridge Regression. Specifically, we identified 46 records with salaries exceeding $1,000,000 (peaking at $50M) and 304 records below $100. These anomalies likely represent data entry errors or currency conversion issues rather than genuine market rates. Similarly, we scrutinized the experience-related features for logical consistency. While rare, we detected 54 respondents claiming more than 60 years of work experience and 71 respondents claiming more than 60 years of coding experience. Given the target demographic of the survey, these values are statistically suspicious. Consequently, they will be treated as noise. This leads to the conclusion that the data quality assessment necessitates a rigorous cleaning pipeline, including dropping rows with missing targets, capping or trimming outliers, and imputing missing categorical features.

2.4 Visual Inspection Findings

Visual inspection of the 31 selected features revealed critical distribution patterns necessitating specific handling. Note that these findings have been highly summarized for brevity, and not every identified anomaly is detailed here.The target variable, `ConvertedCompYearly`, exhibits extreme right-skewness with a long tail extending to $50M; unlike the plausible "veteran" tail observed in `WorkExp`, these salary outliers likely represent data artifacts that violate homoscedasticity assumptions. Structurally, technical stack features (e.g., `Language`) display a combinatorial nature requiring multi-label encoding, while demographic fields like `MainBranch` and `Industry` show a strong selection bias towards established professionals in software development. Furthermore, post-pandemic work trends confirm "Remote" and "Hybrid" modes as the statistical norm. Finally, analysis of ordinal variables identified clear hierarchical structures in `OrgSize` and a left-skewed, generally satisfied sentiment in `JobSat`, contrasting with the discrete, spiky distributions of other sentiment scores.

2.5 Outlier Analysis and Handling Strategy

We performed a targeted analysis of the numerical features, specifically the target variable (`ConvertedCompYearly`) and experience metrics (`WorkExp`, `YearsCode`), to identify high-leverage points that could destabilize the linear regression model.

2.5.1 Visual Inspection of Distributions. As already found in Section 2.4 and illustrated in Figure 4, the target variable is characterized by extreme right-skewness. The original distribution (left panel) is dominated by a tight cluster near zero and a long tail extending to $50M, rendering the boxplot interquartile range (IQR) nearly invisible.

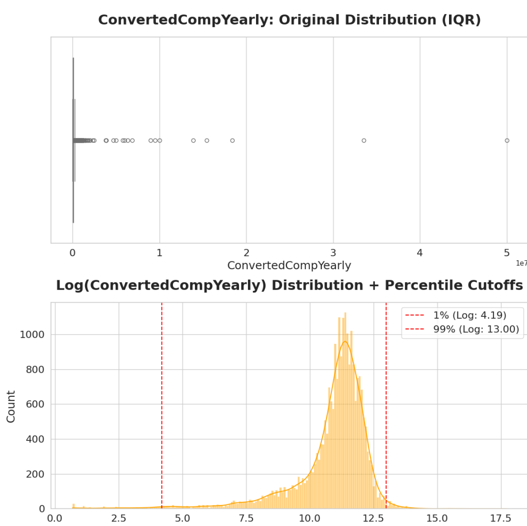

Figure 4: Target Variable Analysis. Left: Original boxplot shows extreme compression due to outliers up to $50M. Right: Log-transformed distribution reveals clearer bell curve, with red lines indicating proposed 1% and 99% capping thresholds.

Similarly, the experience variables (Figures 5 and 6) reveal a "veteran tail." While physically possible, respondents with > 50 years of experience act as statistical outliers in a linear context. As seen in these distributions, the standard IQR method aggressively flags senior professionals (30+ years) as outliers. However, the log-transformed histograms suggest that capping at the 99th percentile is a more data-conserving strategy than strict IQR removal.

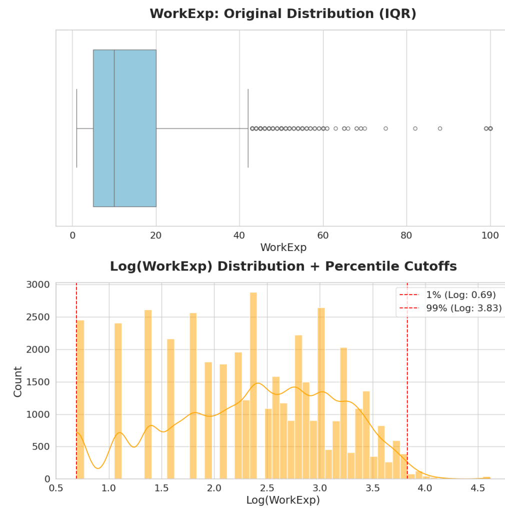

Figure 5: Work Experience Distribution.

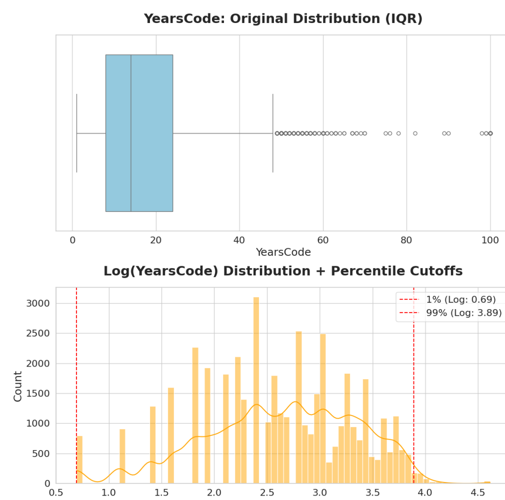

Figure 6: Years Coding Distribution.

2.5.2 Methodological Decision: IQR vs. Percentile Capping. Initial tests using the standard Interquartile Range (IQR) method for outlier removal proved problematic. As seen in the boxplots above, the IQR method classifies a significant portion of valid data (e.g., senior developers with 40+ years of experience) as outliers. Removing these entries would bias the model against high-experience professionals, which lead to the selection of Log-Transformation in combination with 99th Percentile Capping.

2.5.3 Categorical and Ordinal Feature Analysis. Beyond numerical extremes, we analyzed the distribution of categorical and ordinal variables. We identified distinct challenges regarding **Categorical Long Tails**, where features such as `DevType`, `Country`, and `Industry` exhibit high cardinality with rare labels. To prevent overfitting, we will apply **Frequency Thresholding** during the Data Preparation phase to group these rare instances into a generic "Other" category, while the extreme imbalance in `MainBranch` will be addressed via binary grouping. Regarding **Ordinal Validity**, analysis confirmed that low values in `JobSat` represent valid negative sentiment and will be retained; conversely, non-informative labels in other columns (e.g., "I don't know" in `OrgSize`) do not reflect scalar rank and will be flagged for exclusion. Finally, for **Ranked Features** such as `TechEndorse`, prior correlation filtering has already isolated the statistically significant attributes.

2.6 Ethical Aspects

A significant ethical consideration regarding this model is the suitability of broad survey data for predicting individual compensation. Survey data relies on general statistics and may fail to account for an individual's unique context. Utilizing this model for real-world compensation decisions carries the risk of unfairly capping salaries for employees who do not align with the "average" profile, potentially negatively impacting livelihoods. Furthermore, the data exhibits potential geographic and economic bias. Respondents are clustered into distinct economic zones—for example, those in Ukraine (reporting in UAH) versus Western nations (reporting in USD or EUR). If the model fails to account for cost-of-living disparities, it may unfairly categorize senior developers from certain regions as "low value." Additionally, ageism is a concern; the data aggregates all respondents over the age of 65 into a single category, which may lead the model to underestimate the technological proficiency of older demographics. Finally, a systemic preference for formal education over self-taught skills risks penalizing experienced developers who lack traditional university credentials.

2.7 Risks and Bias

A fundamental risk to the analysis is the veracity of self-reported data. Survey responses may contain exaggerated claims or inaccuracies. If this response bias correlates with specific locations or age groups, it introduces hidden latent biases that are difficult to detect.

Financially, the `CompTotal` variable presents significant data quality risks due to mixed currencies. Respondents report figures ranging from thousands (EUR) to millions (UAH) without standardization. Due to this lack of comparability, `CompTotal` is excluded from the final dataset. The analysis will instead rely on `ConvertedCompYearly`. However, this variable contains missing values for a significant portion of the user base (e.g., students), resulting in data gaps. Subjective features also pose a risk; for instance, a respondent's sentiment regarding AI could skew self-reported productivity metrics.

2.8 Action Plan

The data preparation strategy prioritizes regression validity by removing rows with missing target values and eliminating redundant currency columns (`CompTotal`, `Currency`) to prevent multicollinearity.

(1) **Target Cleaning:** Rows with missing `ConvertedCompYearly` values are removed to ensure a stable ground truth.
(2) **Outlier Handling:** We apply a log-transformation followed by 99^{th} percentile capping to `ConvertedCompYearly`, `WorkExp`, and `YearsCode`, managing skewness while retaining valid veteran data.
(3) **Imputation:** Missing `WorkExp` is imputed via age-group medians; missing `TechEndorse` values are zero-filled (indicating no endorsement).
(4) **Ranking Inversion:** Ranking columns (e.g., `TechEndorse`) are inverted so that higher numerical values reflect stronger endorsement.
(5) **Feature Encoding:** Semicolon-delimited strings (e.g., `Language`) are multi-hot encoded, ordinal ranges are mapped to sequential numeric values, and high-cardinality nominals (e.g., `Country`) are reduced via Top-N frequency thresholding.
(6) **Selection & Scaling:** Non-informative responses (e.g., "Prefer not to say") are removed, and all numeric features are scaled to normalize model weights.

3 Data Preparation

3.1 Missing values

Rows with missing values in the target variable ConvertedCompYearly were removed, since a valid salary label is required for our regression model and such records cannot be used for training or evaluation. Further, using mean values was also not the way to go for us as it creates untrue / artificial values. For the two key experience-related predictors, WorkExp and YearsCode, missing values were imputed using the median of each column. Median imputation was chosen because experience-related values can vary strongly and sometimes contain very large values, which would distort the mean.

3.2 Handle Outliers

Regarding outliers, after the Data Understanding phase we thought it would be to good to remove outliers of the following features: ConvertedCompYearly, WorkExp and YearsCode. More information can be seen in section 2.5 Initially, we wanted to remove the 99th percentile. Afterwards, during the Evaluation phase we also played around and after some iterations we found out, removing the 95th percentile brings a good result and everything below it does not really contribute that much anymore. We did the same iterating steps for WorkExp and YearsCode where the values below the 1st and above the 99th percentile have then be removed.

3.3 Inverting Ranking

Initially there have been a lot of columns regarding TechEndorse and JobSatPoints. The within the Data Understanding phase most important one have been kept, the other ones deleted. Note: The * at the end is a placeholder for different numbers: **TechEndors_*, JobSatPoints_* JobSat_*.** The problem is, that giving a 1 here was meant that it is the best possible ranking a developer could give. But as our regressions model should take higher values as "more" or "better" it was clear that it is necessary to invert these features.

3.4 Ordinal Encoding

Several survey questions were originally represented as categorical text labels rather than numeric values, necessitating ordinal encoding. The specific columns processed included **Age**, **Education**, **Organization size**, **AI usage**, and **Feelings towards AI**. Within each feature, categories were mapped to sequential integers where higher values correspond to a "higher" or "better" rank (e.g., assigning a higher numerical value to a Bachelor's degree compared to primary education).

3.5 Multi-Select Column Transformation

Certain questions allowed respondents to select multiple answers, resulting in semi-structured string data (e.g., "Python;Java;C++"). This structure affected the **Programming Languages**, **Databases**, **Cloud Platforms**, **Web Frameworks**, **Development Environments**, **Operating Systems**, and **Additional Employments** columns. To address this, we applied a 1-to-n encoding strategy (multi-label binarization). For every unique answer option within a feature, a distinct binary column was created, assigned a value of 1 if the option was selected by the respondent and 0 otherwise.

3.6 Categorical columns transform

Similar to the Multi-Select columns some form of 1-to-n encoding was used regarding the Categorical features as well. Plain 1-to-n encoding was used for the following features: **Main Branches (where a developer works in)**, **Employment status**, **Development Types**, **Industries**, and **Remote Work**. Regarding the Development types it is important to note, that we considered some of the types not relevant for our model and therefore deleted these rows. The decision was completely subjective made by student b regarding our use case.
Regarding the country a developer works in Top-N encoding was used, as using all possible countries as an own column would have blown up the dataset to much. Top 19 countries were chosen in order to their own column and the rest of the Countries was summarized within an "Other" column.

3.7 Removal of not needed columns

This step initially deleted all columns which were processed and not needed anymore like the ordinal and categorical feature columns. Later on when it came to the modelling part of student a we realized that some columns which already should have deleted by student a above have not been deleted, we just added the affected columns in this part as well in order to make sure that they are always properly deleted when going through the notebook.

3.8 Scaling

After the initial modeling and evaluation part we didn't have this part of the data preparation, but after my team partner and we had a little chat we figured, that we missed out on scaling certain columns and that implementing it would help us in reach better results with our model. So, after again playing around with scaling algorithms the MinMaxScaler was used for the numerical columns where the value was inverted and for the ordinal mapped values. Further, the continuous variables WorkExp and YearsCode were transformed using a Yeo–Johnson power transformation.[6]

3.9 Preprocessing steps considered but not applied

3.9.1 Removing Outliers. Regarding the removal of outliers we had two separate approaches in order to handle them, one being the removal of every row that has a salary above 500.000 per year and the second one being the removal with the usage of percentiles.

In the end we decided against taking absolute number as a border for removal, because it is in our opinion not wise to do that. Having absolute numbers as a salary cap me be working for now, but in the future when salaries generally rise, the border of 500.000 might be outdated and therefore could lead us to wrong decisions. With using perceptiles on the other hand, you always go with the relative flow of the numbers.

3.9.2 Binning. We did not use binning at all (apart of the already "binned" columns like age) because with binning for example the salaries of the developers we would have created classes which afterwards would have to be encoded again in order to be usable for our regression problem. And as this would have been a lot of work for no real benefit we did not do that.

3.9.3 Attribute removal. Regarding attribute removal we had a long discussion about which attributes to use and which to remove from our dataset, always debating whether these attributes would have contributed something to our problem / model or not. Especially regarding the attribute "RemoteWork", which basically listed all types of working (from fully remote to fully in the office) we had a long discussion about whether this attribute would be interesting or not. After a bit of searching through the internet we then decided to keep the attribute and therefore did not apply the step of attribute removal.

3.9.4 One-Hot Encoding for countries. We decided against doing one-Hot encoding for every country as it would have resulted in more than 150 columns to cover each country. We therefore decided to analyze how many countries developers are withing the Top 5, 10, 15, and 20 and then decided to go with Top-N-Encoding.

3.9.5 Textual fields. We considered to go through and analyze all columns which consisted of textual answers that were given by the developers, but decided against it, as it would have been far to much to analyze and also difficult to properly encode it with the right values. Therefore we decided to drop all these attributes which resulted in having less noisy features within our dataset.

3.10 Derived attributes

3.10.1 Countries –> Regions. One possible option for making a derived attributes would have been to group countries to regions / continents. But as there are a lot of differences betweend for example western and eastern europe, we decided against it.

3.10.2 Work experience in years –> seniority level. We could have created a column and classify developers based on their working experience as junioer, senior, etc. But as the Working experience is a numerical input anyway and also is "better" the higher the value is, it didn't make much sense for us. We further think, that with kind of grouping developers based on their work experience, we might create a bias which we do not want.

3.10.3 Job satisfactory. Regarding the job satisfactory we also could have created derived attributes that for example say if you gave your job between 10-14 points yout job is satisfying, but with our regression problem we would have needed to encode it again as not satisfying for example would have gotten a 1 and satisfying a 2. This therefore wouldn't have made any sense as well.

3.10.4 1-to-n coding. The only real thing we did regarding the deriving of attributes was to make use of the 1-to-n encoding. As we had a lot of attributes that were either ordinal, Multi-Select or categorical, this was in our opinion the best way in order to transform the values of the affected attributes to make our dataset ready for the modeling.

3.11 External Data Sources

Below now follow three potential datasets that we could have used in order to train, validate and test our model.
From Kaggle [16]: This could have been a good alternative to our chosen dataset, but this one only has 14 attributes and in the specification it would have been at least 15, so we wouldn't have been able to use this dataset anyways. Furthermore, this dataset does not have any attributes, that our chosen dataset wouldn't have anyway.
From back4app [2]: While doing the research we stumbled across this dataset right here, but in my opinion we wouldn't be able to derive some high quality model from this dataset.
From arc.dev [1]: If we wanted to dive deeper into the salaries regarding developers that are working remote, this dataset would have actually been interesting for us.
From Datarade [4]: This datasets also look very good on the first sight, but we would have to be very careful on using one of these datasets, because some of them only cover one country like USA. This would have created a population bias which we do not want.

4 Modeling

4.1 Algorithm Selection

Consistent with the objectives defined in the Data Understanding phase, we aimed for a linear approach to establish a solid baseline. We selected **Ridge Regression (L2 Regularization)** as the primary modeling technique. This and the following section thereby reference the Ridge Regression Model as defined by the sklearn library in python[14].

4.1.1 Justification. Our dataset is characterized by a moderate row count ($N \approx 25,000$) but high dimensionality ($D \approx 400$ features) resulting from the extensive one-hot encoding of multi-select and categorical variables.

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Standard Ordinary Least Squares (OLS) regression is highly vulnerable to overfitting in this setting due to multicollinearity (high correlation between features). McDonald highlights that in such “ill-conditioned” problems, OLS coefficient estimates become unstable and exhibit high variance, making them unreliable for prediction [7, 10]. Ridge Regression addresses this by adding an L2 penalty to the loss function:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \alpha \sum \beta_j^2$$

This penalty shrinks coefficients (β) towards zero, effectively trading a small amount of bias for a significant reduction in variance [10], which prevents overfitting without eliminating variables entirely.

4.2.2 Alternative Algorithms Considered. We evaluated several alternative approaches before finalizing our choice. **Lasso Regression (L1)** was considered for its feature selection capabilities, but it can be unstable when features are highly correlated, arbitrarily selecting one variable while reducing others to zero; in contrast, Ridge is generally more robust in preserving signal from correlated groups [11]. We also examined **ElasticNet**, which combines L1 and L2 penalties, but decided against it as it introduces an additional hyperparameter (`l1_ratio`) to tune, thereby increasing computational complexity for this baseline phase [9]. Finally, while **Histogram-based Gradient Boosting (HGBC)** is highly efficient for tabular data and handles non-linear relationships better than linear models, we prioritized Ridge to establish a high-interpretability baseline where feature importance can be directly analyzed via coefficient values [8].

4.3 Hyperparameter Selection

4.3.1 Tuning Strategy: Regularization Strength (α). The primary hyperparameter selected for optimization is α (alpha). In Ridge Regression, this parameter controls the regularization strength, governing the trade-off between model complexity and error.
Justification: A value of $\alpha = 0$ is equivalent to standard OLS (high variance), while very high α values lead to underfitting (high bias). Given the high dimensionality of the dataset, optimizing α is critical to effectively dampen noise without losing predictive signal. We employ a logarithmic grid search to identify the optimal magnitude.

4.3.2 Final Configuration. The complete configuration used for the training phase is summarized in the table 1 below, which are the default values provided by sklearn [14]:

Table 1: Hyperparameter Configuration for Ridge Regression

Parameter	Description	Strategy/Value
`alpha`	Regularization strength (L2 penalty)	**Tuned** (Log Grid)
`solver`	Optimization algorithm	`'auto'`
`fit_intercept`	Calculates model intercept	`True`
`tol`	Convergence tolerance	1×10^{-4}
`positive`	Force positive coefficients	`False`
`random_state`	Seed for reproducibility	42

4.4 Splitting Strategy

We employed a 70% Training, 15% Validation, and 15% Testing split. A random state of 42 was used to ensure reproducibility. Since this is a regression task on continuous salary data, standard stratification was not applied, but distribution checks (mean/std) confirm that the splits are statistically representative of the overall population.

4.4.1 Distribution Consistency. To ensure the model is evaluated on data that closely resembles the training conditions, we analyzed the distribution of the target variable (`ConvertedCompYearly`) across all splits. Figure 7 displays the Kernel Density Estimate (KDE), showing that the training, validation, and testing curves overlap almost perfectly. This confirms that no specific salary range is over- or under-represented.

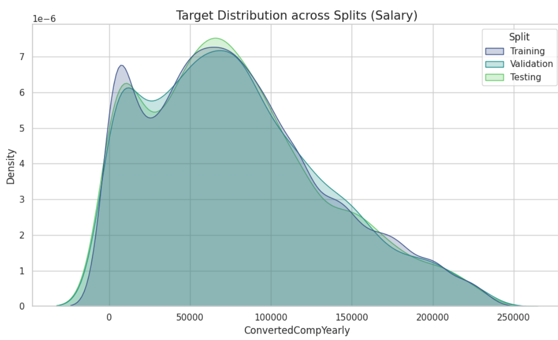

Figure 7: Target Distribution across Splits (Salary). The overlapping curves indicate a statistically representative split.

This consistency was further validated through boxplot analysis, which demonstrated that the Median and Interquartile Ranges (IQR) for salary remain stable across all partitions. We extended this verification to key features to ensure the model learns from a balanced demographic. For instance, the normalized distribution of `WorkExp` showed consistent density across splits, ensuring that both junior and senior profiles are equally distributed in the hold-out sets.

4.5 Training & Tuning Results

We optimized the regularization strength by evaluating the model across a logarithmic grid $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$. Table 2 summarizes the performance metrics for each configuration.

Table 2: Grid Search Results: Regularization Impact

Alpha (α)	R^2 Score	MAE (USD)
0.01	0.5712	26,325.67
0.1	0.5712	26,325.48
1.0	0.5712	26,323.76
10.0	**0.5713**	**26,313.32**
100.0	0.5662	26,500.52

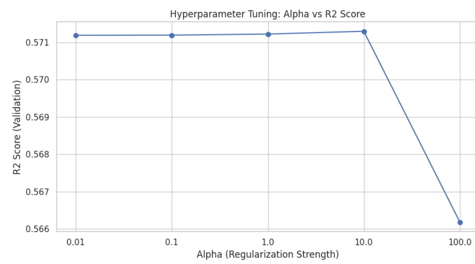

Figure 8: Hyperparameter Tuning Results: Validation R^2 scores across Alpha values.

As visually confirmed in Figure 8, the model exhibits high stability at lower regularization levels ($\alpha \leq 1.0$) before reaching a peak performance at $\alpha = 10.0$. This specific value represents the optimal trade-off ("sweet spot"), where the L2 penalty is sufficient to dampen noise—reducing the Mean Absolute Error to its minimum of $26,313—without suppressing relevant features. Beyond this point, at $\alpha = 100.0$, the curve shows a sharp degradation in accuracy (R^2 drops to 0.5662), indicating that the regularization has become too dominant and forced the model into an underfitted state.

4.6 Final Model Selection

Based on the validation results, the hyperparameter configuration with $\alpha = 10.0$ was selected as the optimal model.

Table 3: Final Model Validation Performance

Metric	Value	Interpretation
R^2 Score	0.5713	Explains ∼57.1% of variance in salary data.
MAE	$26,313.32	Average prediction error in USD.

An important note is that the metrics reported below represent performance on the **Validation Set (15%)** only. The held-out Test Set remains completely unseen at this stage to ensure it serves as an unbiased evaluator for the final production model.

4.7 Retraining Process

Following the selection of the optimal hyperparameter ($\alpha = 10.0$), the final production model was not merely the specific artifact from the validation run, but rather the result of a complete retraining step.

To maximize the learning capacity of the model, we merged the Training (70%) and Validation (15%) sets to form a larger training corpus representing 85% of the total data. A fresh Ridge Regressor was then initialized with the selected $\alpha = 10.0$ and fitted on this combined dataset. This approach allows the model to leverage the maximum amount of available data to learn underlying patterns, thereby improving its potential for generalization on the unseen Test set (the final 15%) without violating the hold-out principle.

4.8 Non-Linear Validation (HGB)

To assess potential gains from non-linear modeling, we tracked an exploratory `HistGradientBoostingRegressor` experiment in the provenance graph, varying regularization and tree depth. As shown in Table 4, the best configuration achieved a validation R^2 of **0.5957**, outperforming the Ridge baseline (0.5713) by approximately 2.4%. While this confirms the presence of non-linear interactions, we retain Ridge Regression as the primary production model for its superior interpretability, though future CRISP-DM cycles should investigate tree-based approaches further.

Table 4: HGB Benchmarking Results

L2 Regularization	Max Depth	Train R^2	Validation R^2
0.0 (Baseline)	None	0.7093	**0.5957**
1.0 (Moderate)	10	0.7102	0.5955
10.0 (Strong)	5	0.6626	0.5787

5 Evaluation

5.1 Performance reflection

To evaluate the model performance, we calculated the coefficient of determination and the mean absolute error. Our initial results were not satisfactory, so we performed several iterations of data preparation, modeling, and evaluation. At first, the R^2 was only about 50%. After improving the data, the linear model reached nearly 60%. Since this was still not fully satisfying, a non-linear model was developed, which achieved up to 70% on the training data. On the test set, the non-linear model performed slightly better with an R^2 of 60.35%, compared to 59.15% for the linear model, which can be considered acceptable. However, the mean absolute error of 25,220.51 is still too high.

5.2 State-of-the-art performance

Regarding our dataset, we identified three websites that documented the use of the same dataset as ours. But please note that because our dataset is relatively new, all three models were trained and tested using developer survey data from the previous year. Although this is generally not considered best practice, it was the most feasible approach for us. The first model was found on Kaggle [5], where only the root mean squared error was reported. The second model was published on Medium [12]. This model achieved an R^2 of 0.045 and a root mean squared error of 113,000. The final model was found on GitHub, [15] where three performance metrics were reported: an R^2 of 0.619, a mean absolute error of 25,121.2, and a root mean squared error of 37,068.77.

5.3 Baseline Performance

To establish a rigorous trivial acceptor and baseline performance, we calculated the standard metrics, yielding an R^2 of **-0.0005**, a Mean Absolute Error (MAE) of **44,279.54**, and a Root Mean Square Error (RMSE) of **54,765.98**. These baseline calculations were derived using methodologies referenced in [3].

5.4 Comparing to State-of-the-art and baseline

5.4.1 Baseline performance. Overall, our model was able to closely match or even beat the reported performance metrics of comparable models found online. Starting with the baseline / trivial acceptor, our model unsurprisingly outperformed all baseline metrics, with none of the baseline results coming close to our model's performance.

5.4.2 State-of-the-art performance. With an RMSE of 33.2k USD on the test set, the Kaggle model performed slightly better than our model, which achieved an RMSE of approximately 34.5k EUR. After comparing our calculated metrics with those reported in the Medium article, it became evident that the Medium model performs considerably worse. Consequently, our model clearly outperformed all of their reported metrics. When comparing our model to the GitHub implementation, our model achieved a lower RMSEand performed slightly worse in terms of the R^2 score and the MAE. In conclusion, our model can be considered close to state-of-the-art performance for this task. None of the compared models significantly outperform our model, and even in cases where marginally better results are achieved, our performance remains very close to the reported benchmarks.

5.4.3 Regression errors. Our model has regression errors in certain parts of the data space, which occur the higher a salary gets. Based on the diagram in 3.2 we subjectively created three salary classes:

Class	Lower Border	Upper Border
Low	0	50 000
Medium	50 000	150 000
High	150 000	max_salary

Table 5: Salary class borders used for regression error analysis

The lowest regression errors have been recorded within the medium class, while we had more regression error in the Low class and the most errors within the hightest class. The following diagram should help to better visualize this:

5.5 Comparison of the performance against success criteria

This section evaluates the model against the technical data mining success criteria defined at the project's inception, while the comparison against business-specific objectives is addressed in the Deployment section. The technical targets included achieving an $R^2 > 80\%$, maintaining a Mean Absolute Error (MAE) below 3,000 EUR, ensuring less than 10% missing data, and guaranteeing readability for the HR department. The evaluation yields mixed results regarding the quantitative performance metrics. The target of achieving an $R^2 > 80\%$ proved overly ambitious given the inherent noise in salary data. However, comparative analysis indicates

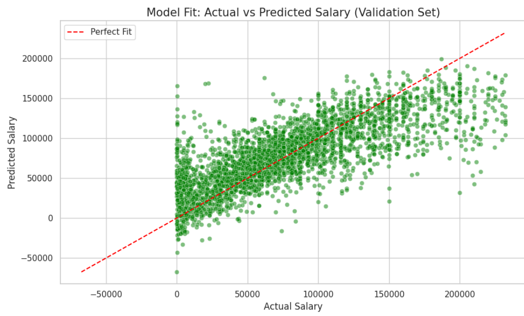

Figure 9: Model Fit: Actual vs Predicted.

that the developed model aligns with or exceeds similar external benchmarks, suggesting that the initial threshold was unrealistic for this specific domain. Similarly, the goal of maintaining a MAE below 3,000 was not met, with the final error margin exceeding 20,000. While numerically high, this performance is contextually acceptable given the high variance of the target variable and the ambitious nature of the original goal.

In contrast, the data quality objective was fully achieved, as the data preparation phase successfully eliminated all missing values, resulting in 0% missingness. Finally, the criterion regarding readability for the HR department remains technically unverified at this stage. As readability is a subjective metric dependent on end-user experience, it cannot be definitively assessed by the development team and requires external validation.

5.6 Protected attribute

Regarding our dataset, there there are several protected attributes like the country a developer is working in or the type of working (in-person, remote, hybrid, ...). For this task we decided to evaluate if the model exhibits a bias towards the country of a developer, which is the case for just a small amount of countries like the USA, Romania and Belgium that are therefore driving up the RMSE as well as the MAE.

6 Deployment

6.1 Comparison And Recommendations for Business Objectives and Success criteria

6.1.1 Business success criteria. **HR contract review:** The model allows HR to quickly see salary differences, so this success criterion is met.
Reducing staff turnover: The model helps standardizing salaries, improving employee morale and reducing terminations. This criterion is met as well.
Offer acceptance: The model supports more reasonable and competitive offers, but HR cannot fully rely on it. This criterion is only partly met.
Budget optimization: The model helps avoid over- or underpaying, but outliers are possible and salaries still need validation. We therefore do not know if this criterion is met.
Faster time-to-hire: We need an empirical study from HR to evaluate this. At the moment, we cannot tell if this criterion is met.

6.1.2 Business objectives met. **Identifying significant salaries:** The model makes this possible, but reliable job satisfaction data may be hard to obtain in a non-anonymous way.
Standardizing salary offers: Time will tell if we fully achieved this, but we believe the criterion is fulfilled.
Company budget: The model improves salary predictions, allowing the CFO to plan the budget with higher reliance.

6.1.3 Form of deployment. Fully automating the deployment would not be wise. A hybrid solution is better, where the model helps in finding a decision but never does so alone due to bias concerns. There should always be a human in the loop monitoring everything.

6.1.4 Recommendations for subsequent analysis. The best way to analyze predictions is to use logs that give all possible information, especially for cases where the model predicted nonsense. HR feedback should also be considered, as they are the main users.
This may not improve accuracy, but it can improve how the model is handled. We should also conduct internal AI-audits from time to time to check whether the model works and is used as intended and to find ways of improving it. Logging and feedback are important parts of such audits.

6.2 Ethical Aspects

6.2.1 Ethical aspects in Business Understanding. This section addresses specific risks identified during the initial project phase. The absence of gender data in the source inherently minimizes the risk of **Gender-based bias**. While missing values were resolved, residual **Noisy Data** remains a risk that could impair decision accuracy. A critical trade-off exists regarding **Geography**; while including country introduces location bias, excluding it ignores essential economic variables (taxation, cost of living) required for valid prediction. Furthermore, **Sampling Bias** in the survey population may underrepresent certain demographic groups. Potential misalignment with current **Fritz & Maier AG salaries** is considered a low priority, as the company specifically aims to modernize compensation. Finally, **Governance** is ensured via comprehensive logging, feedback loops, and internal audits.

6.2.2 Ethical aspects in Data Understanding. Regarding **Dataset Suitability**, the survey serves as a cost-effective proxy for real-world data, though it carries inherent limitations. Reliance on this data without oversight could lead to unfair offers, negatively impacting time-to-hire and retention. To mitigate risks of **Unfair Treatment**—specifically regarding ageism or educational bias—human-in-the-loop validation is mandatory. We propose a two-tier review process: HR must validate general alignment to prevent demographic discrimination, while technical supervisors should verify that the predicted salary corresponds to the applicant's actual skill level rather than solely their formal credentials.

6.3 Monitoring Plan

6.3.1 Data Drift & Concept Drift. Salary data is intrinsically temporal. A model trained on 2025 data will likely underestimate salaries in 2026 due to inflation. To mitigate this, we propose a quarterly "Index Adjustment" where model predictions are validated against current market rates (using fresh scraped data or CPI indices) to detect if the baseline is drifting. Furthermore, the technology landscape evolves rapidly. We must continuously monitor the input distribution of the "Skills" and "DevType" features; if the frequency of a new skill (e.g., "Mojo" or "Q#") spikes in new applications but is absent or rare in our training distribution, the model will be flagged for retraining.

6.3.2 Performance Monitoring (Human-in-the-Loop). As this model is designed as a decision-support tool rather than an automated agent, we will implement a "Prediction vs. Actual" log. HR personnel will record the final signed offer alongside the model's initial prediction. If the Mean Absolute Error (MAE) of the deployed model exceeds the Test Set MAE by more than 10% over a rolling 3-month period, a retraining alert is triggered. Additionally, we will actively monitor the rate of "Outlier Flags" generated by the deployment pipeline. If the model flags more than 20% of new applicants as outliers (indicating salary expectations are systematically too high or low), it suggests the underlying data distribution has shifted significantly compared to the 2025 survey baseline.

6.3.3 Bias Auditing. Finally, to ensure ethical fairness, we will execute the bias evaluation script (established in the Evaluation phase) on newly accumulated applicant data annually. This ensures the model does not develop or amplify bias against specific geographic regions or non-traditional educational backgrounds over time.

6.4 Reflection on Reproducibility

6.4.1 Strengths via Provenance & Environment. The project exhibits robust reproducibility through several mechanisms. First, by logging every activity (e.g., `:handle_outliers`, `:scale_columns`) and entity into the Knowledge Graph, we have created a semantic trail. This allows auditors to trace exactly which transformation steps led to the final model, exceeding standard code documentation by making the process machine-queryable. Second, we ensured determinism by utilizing `random_state=42` for all stochastic processes, ensuring that re-running the notebook yields the exact same coefficients and R^2 scores given the same input. Finally, the use of a dedicated Conda environment (defined via `requirements.txt`) ensures that library version discrepancies—such as Scikit-Learn updates altering default solver behaviors—do not alter the results.

6.4.2 Limitations & Challenges. Despite these measures, certain limitations persist. While the code execution is reproducible, the *rationale* behind specific manual decisions—such as the choice of outlier thresholds or the exclusion of columns like `RemoteWork`—remains subjective. While the graph documents *what* happened, reproducing the *thought process* requires this accompanying report. Additionally, the project relies on the external Stack Overflow 2025 CSV file; if this source file is modified or removed from the data directory, the graph pointers will break. True reproducibility would require version-controlling the raw data artifact (e.g., using DVC) alongside the code. Finally, execution times logged in the graph are hardware-dependent, meaning that efficiency metrics cannot be guaranteed across different machines.

7 Conclusion

In this project, we successfully applied the CRISP-DM methodology to the Stack Overflow 2025 Annual Developer Survey to develop a salary benchmarking tool for Fritz & Maier AG. The primary business objective was to establish a data-driven model capable of identifying internal pay inequities and standardizing offers for new hires. Through iterative cycles of data understanding and preparation, we developed a linear baseline model using **Ridge Regression**, which achieved an R^2 of 0.5713 and a Mean Absolute Error (MAE) of $26,313 on the validation set. To validate these results against non-linear complexities, we trained a comparative **Histogram-based Gradient Boosting (HGB)** model, which improved performance to a Test R^2 of 60.35%. Our analysis indicates that while the model outperformed the trivial baseline and achieved results comparable to state-of-the-art benchmarks found on Kaggle and GitHub, precise salary prediction remains challenging due to the extreme noise and right-skewness of the data. Specifically, we observed that regression errors increase with higher salary brackets; while the model performs adequately for low-to-medium salary ranges, it struggles with the highest bracket (> $150,000). Consequently, regarding business viability, the model is best deployed as a decision-support tool rather than an autonomous agent. While it successfully meets the criterion for reviewing current contracts and reducing turnover risk, it requires human supervision for budget optimization to mitigate the impact of potential outliers.

7.1 Lessons Learned

Throughout the lifecycle of this project, we encountered several challenges that provided valuable learning opportunities regarding the data mining process. A significant methodological lesson involved the dependency between *Data Understanding* and *Data Preparation*. We found that certain exploratory analyses were only meaningful *after* specific preprocessing steps had been applied. However, strictly adhering to the initial project plan sometimes caused friction, as we had to partially transform data before fully analyzing it. This reinforced the non-linear, cyclic nature of CRISP-DM, where moving back and forth between phases was necessary to derive valid insights. Additionally, we learned that standard statistical techniques must be adapted to the domain. For instance, using the standard Interquartile Range (IQR) method for outlier removal proved destructive, as it flagged experienced professionals (40+ years of tenure) as outliers. Adopting a log-transformation combined with percentile capping allowed us to retain these valuable signals while stabilizing the model.

7.2 Feedback on the Exercise

Reflecting on the course structure and the assignment, we would like to offer the following feedback:

Process and Instructions: We noticed discrepancies between the specific assignment instructions and the standard CRISP-DM process definition. This sometimes led to confusions for us.

Documentation Workflow: We found the workflow of generating the report directly from Python comments to be problematic as our provenance comments didn't fit the style of the report and therefore we had to change a lot manually.

Course Content: Finally, while classification is covered well, we believe the curriculum would benefit from a dedicated section on **Regression Analysis**.

8 AI-Disclaimer

As discussed in the lecture with Professor Rauber, we used Generative AI as a helper throughout this project to support several different tasks, beginning with the brainstorming phase and the clarification of specific assignment requirements. It provided structured feedback on our adherence to the CRISP-DM process and served as a knowledge base for exploring fundamental concepts within Regression Machine Learning. For the technical part AI helped us choose the right algorithms, fix bugs in our code, and log our steps into the knowledge graph. Finally, we used it to improve the wording and make the text clearer in both this report and the Provenance Ontology documentation.

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